Trigonometry & Additional Topics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometry**

In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Adjacent $= \sqrt{20^2 - 12^2} = \sqrt{400 - 144} = \sqrt{256} = 16$. $\sin\theta = \frac{12}{20} = \frac{3}{5}$. $\cos\theta = \frac{16}{20} = \frac{4}{5}$. $\tan\theta = \frac{12}{16} = \frac{3}{4}$.

2 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

Opposite $= \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\sin\theta = \frac{24}{25}$. $\tan\theta = \frac{24}{7}$.

3 **[Foundational]** • **Target: Right-triangle trigonometry**

A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

Height $= 15\sin(50^\circ) \approx 15(0.766) \approx 11.5$ feet.

4 **[Foundational]** • **Target: The Pythagorean identity**

Given $\sin\theta = \frac{5}{13}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\cos\theta = \frac{12}{13}$.

5 **[Foundational]** • **Target: The Pythagorean identity**

Given $\cos\theta = \frac{8}{17}$ with θ acute, find $\sin\theta$.

$\sin^2\theta = 1 - \frac{64}{289} = \frac{225}{289}$. Since θ is acute, $\sin\theta = \frac{15}{17}$.

6 **[Foundational]** • **Target: Exact trigonometric values**

State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.

$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

7 **[Foundational]** • **Target: Solving for missing sides and angles**

A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32°, as shown below. Find the height of the tower, to the nearest foot.

$h = 150\tan(32^\circ) \approx 150(0.6249) \approx 94$ feet.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Solving for missing sides and angles**

From the top of a 60-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

The angle of depression equals the angle of elevation from the boat, 25°, so $\tan(25^\circ) = \frac{60}{d}$, giving $d = \frac{60}{\tan(25^\circ)} \approx \frac{60}{0.4663} \approx 129$ feet.

9 **[Intermediate]** • **Target: Complex numbers: arithmetic**

Simplify i^7, using $i^2 = -1$.

$i^7 = i^4 \cdot i^3 = 1 \cdot i^3 = i^2 \cdot i = -1 \cdot i = -i$.

10 **[Intermediate]** • **Target: Complex numbers: modulus and conjugates**

Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

Conjugate: $-3 - 5i$. Product $= (-3 + 5i)(-3 - 5i) = 9 - 25i^2 = 9 + 25 = 34$.

11 **[Intermediate]** • **Target: Vectors: magnitude and operations**

Vector $v = (3, 4)$. Find its magnitude.

$|v| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

12 **[Intermediate]** • **Target: Vectors: magnitude and operations**

Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.

$u + v = (2 + (-6), -5 + 3) = (-4, -2)$.

13 **[Intermediate]** • **Target: Vectors: magnitude and operations**

Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

$2w = (16, -12)$. $|2w| = \sqrt{16^2 + 12^2} = \sqrt{256 + 144} = \sqrt{400} = 20$.

14 **[Intermediate]** • **Target: Radian measure**

Convert 150° to radians, in terms of π.

$150 \times \frac{\pi}{180} = \frac{5\pi}{6}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Radian measure • Cross-Lesson: 8.01 Exponentials & Logarithms

Convert $\frac{7\pi}{4}$ radians to degrees.

$\frac{7\pi}{4} \times \frac{180}{\pi} = 7 \times 45^\circ = 315^\circ$.

16 [Challenging] • Target: A well-designed word problem: measuring the width of a river • Cross-Lesson: 1.01 Heart of Algebra

This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C. (a) At point C, the angle between the bank and the line of sight to the tree is 35°. Use the tangent ratio to find AB, the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24°. Use this second measurement to find a second estimate of AB, to the nearest meter, and state which measurement gives the larger estimate.

(a) $\tan(35^\circ) = \frac{AB}{80}$, so $AB = 80\tan(35^\circ) \approx 80(0.7002) \approx 56$ meters. (b) $CB = \frac{80}{\cos(35^\circ)} \approx \frac{80}{0.8192} \approx 98$ meters. (c) $AB = 130\tan(24^\circ) \approx 130(0.4452) \approx 58$ meters. The second measurement (58 m) is slightly larger than the first (56 m); the two estimates are reasonably consistent.

17 [Challenging] • Target: Radian measure • Cross-Lesson: 2.01 Quadratics & Polynomials

In a right triangle, acute angle θ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

Step 1: In a right triangle with opposite side 5 and hypotenuse 13, the adjacent side is: $\text{adj} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. Step 2: Compute trigonometric ratios:$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \mathbf{\frac{12}{13}}$. $\tan\theta = \frac{\text{opp}}{\text{adj}} = \mathbf{\frac{5}{12}}$.$\sec\theta = \frac{1}{\cos\theta} = \mathbf{\frac{13}{12}}$.

18 [Challenging] • Target: A well-designed word problem: measuring the width of a river • Cross-Lesson: 3.01 Functions & Graphs

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.

Step 1: Convert to degrees: $\theta = \frac{7\pi}{6} \times \frac{180^\circ}{\pi} = 7 \times 30^\circ = \mathbf{210^\circ}$. Step 2: Terminal coordinates in Quadrant III: The reference angle is $210^\circ - 180^\circ = 30^\circ$. In Quadrant III, both coordinates are negative: $x = \cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$, $y = \sin(210^\circ) = -\sin(30^\circ) = -\frac{1}{2}$. The terminal point is $\mathbf{\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)}$.

19 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 4.01 Ratios, Rates & Data

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

Step 1: Expand the numerator: $(3-2i)^2 = 3^2 - 2(3)(2i) + (2i)^2 = 9 - 12i + 4(-1) = 5 - 12i$. Step 2: Multiply by conjugate $1 - i$: $\frac{5-12i}{1+i} = \frac{(5-12i)(1-i)}{(1+i)(1-i)} = \frac{5-5i-12i+12i^2}{1-i^2} = \frac{5-17i-12}{1-(-1)} = \frac{-7-17i}{2}$. In standard form: $\mathbf{-\frac{7}{2} - \frac{17}{2}i}$.

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20 [Challenging] • Target: The Pythagorean identity • Cross-Lesson: 5.01 Statistics & Probability

If $\cos(2x + 10^\circ) = \sin(3x - 15^\circ)$, find the value of x and evaluate $\tan(x + 16^\circ)$.

Step 1: Apply cofunction identity $\cos A = \sin B \Leftrightarrow A + B = 90^\circ$:

$(2x + 10^\circ) + (3x - 15^\circ) = 90^\circ \Rightarrow 5x - 5^\circ = 90^\circ \Rightarrow 5x = 95^\circ \Rightarrow \mathbf{x = 19^\circ}$. Step 2: Evaluate tangent:

$\tan(19^\circ + 16^\circ) = \tan(35^\circ) \approx \mathbf{0.7002}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, the hypotenuse is 87 and the side opposite angle θ is 60. Find $\sin\theta$.

a. $\frac{21}{29}$ **b.** $\frac{20}{29}$ ✓

c. $\frac{20}{21}$ **d.** $\frac{62}{87}$

$\sin\theta = \frac{60}{87} = \frac{20}{29}$.

22 [Foundational] • Target: The Pythagorean identity

In a right triangle, the side opposite angle θ is 6 and the side adjacent is 8. Find $\tan\theta$.

a. 1 **b.** $\frac{4}{5}$

c. $\frac{3}{5}$ **d.** $\frac{3}{4}$ ✓

$\tan\theta = \frac{6}{8} = \frac{3}{4}$.

23 [Foundational] • Target: Exact trigonometric values

In a right triangle, the hypotenuse is 25 and the side adjacent to angle θ is 24. Find $\cos\theta$.

a. $\frac{26}{25}$ **b.** $\frac{24}{25}$ ✓

c. $\frac{7}{24}$ **d.** $\frac{7}{25}$

$\cos\theta = \frac{24}{25}$.

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24 [Foundational] • Target: Solving for missing sides and angles

In a right triangle, angle A and angle B are acute angles such that $\sin(A) = \frac{7}{11}$. What is the value of $\cos(B)$?

a. $\frac{7}{11}$ ✓

b. $\frac{4}{11}$

c. $\frac{8}{11}$

d. $\frac{11}{7}$

In any right triangle, the two acute angles A and B are complementary, meaning $A + B = 90^\circ$ (or $B = 90^\circ - A$). By the cofunction identity: $\cos(B) = \cos(90^\circ - A) = \sin(A)$ Since $\sin(A) = \frac{7}{11}$, we have $\cos(B) = \frac{7}{11}$.

25 [Foundational] • Target: Complex numbers: arithmetic

In the right-angled triangle ABC shown below, $\angle ABC = 90^\circ$, $AB = 3$, and $BC = 4$. What is the length of the hypotenuse AC?

a. **5** ✓

b. 7

c. 12

d. 25

This question involves a direct application of the Pythagorean theorem in a right-angled triangle. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In triangle ABC, with $\angle ABC = 90^\circ$, AB and BC are the two shorter sides (legs), and AC is the hypotenuse. According to the Pythagorean theorem: $AC^2 = AB^2 + BC^2$ Given $AB = 3$ and $BC = 4$: $AC^2 = 3^2 + 4^2$.

26 [Foundational] • Target: Complex numbers: modulus and conjugates

In the right-angled triangle shown, what is the length of side AC if $AB = 10$ and $\angle B = 30^\circ$?

a. **5** ✓

b. 10

c. $5\sqrt{3}$

d. 20

In the right-angled triangle ABC, we are given AB as the hypotenuse and $\angle B = 30^\circ$. We need to find the length of the side opposite to $\angle B$, which is AC. Using the sine function, we have: $\sin(\angle B) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{AC}{AB}$ Substituting the given values: $\sin(30^\circ) = \frac{AC}{10}$ We know that $\sin(30^\circ) = \frac{1}{2}$. So, $\frac{1}{2} = \frac{AC}{10}$. Multiplying both sides by 10 to solve for AC: $AC = 10 \times \frac{1}{2}$.

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27 [Foundational] • Target: Vectors: magnitude and operations

In the right-angled triangle shown below, what is the value of $\sin(\theta)$?

a. $\frac{4}{5}$ ✓ **b.** $\frac{3}{5}$

c. $\frac{5}{3}$ **d.** $\frac{5}{4}$

The sine of an angle in a right-angled triangle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the given triangle, the side opposite to angle θ has a length of 4, and the hypotenuse has a length of 5. Therefore, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{4}{5}$.

28 [Foundational] • Target: Radian measure

In a right triangle, the hypotenuse is 10 and one angle is 30°. What is the length of the side opposite the 30° angle?

a. 6 **b.** **5** ✓

c. 10 **d.** 4

$\sin(30^\circ) = \frac{1}{2} = \frac{opp}{10} \Rightarrow opp = 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: measuring the width of a river

In a right triangle, $\sin\theta = \frac{5}{13}$. What is $\cos\theta$?

a. $\frac{5}{12}$ **b.** $\frac{12}{13}$ ✓

c. $\frac{13}{12}$ **d.** $\frac{5}{13}$

$\cos\theta = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$.

30 [Intermediate] • Target: Right-triangle trigonometry

What is the radian measure of an angle of 330°?

a. $\frac{11\pi}{6}$ ✓ **b.** $\frac{\pi}{33}$

c. $\frac{330\pi}{360}$ **d.** $\frac{330}{180\pi}$

To convert an angle from degrees to radians, multiply by $\frac{\pi}{180^\circ}$: $\text{Radians} = 330^\circ \times \frac{\pi}{180^\circ} = \frac{330\pi}{180} = \frac{11\pi}{6}$.

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31 [Intermediate] • Target: The Pythagorean identity

Given that $i = \sqrt{-1}$, what is the product of the complex numbers $(2 + 4i)(3 - 1i)$?

a. $2 + 10i$

b. $6 - 4i$

c. $10 + 14i$

d. $10 + 10i$ ✓

Expand using the distributive property (FOIL): $(2 + 4i)(3 - 1i) = (2)(3) - 2i + 12i - (4)(1)i^2$ Since $i^2 = -1$: $= 6 + (10)i - (4)(-1) = 6 + 4 + 10i = 10 + 10i$.

32 [Intermediate] • Target: Exact trigonometric values

In a right triangle ABC where angle $C = 90^\circ$, if $\cos(A) = \frac{12}{13}$, what is the value of $\tan(A)$?

a. $\frac{5}{12}$ ✓

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{13}{12}$

Using $\sin^2(A) + \cos^2(A) = 1$, $\sin(A) = \sqrt{1 - (12/13)^2} = \frac{5}{13}$. Then $\tan(A) = \frac{\sin(A)}{\cos(A)} = \frac{5/13}{12/13} = \frac{5}{12}$.

33 [Intermediate] • Target: Solving for missing sides and angles

In a right triangle, the side opposite angle θ is 18, the adjacent side is 24, and the hypotenuse is 30. What is the sine of θ?

a. $\frac{3}{4}$

b. $\frac{5}{3}$

c. $\frac{4}{5}$

d. $\frac{3}{5}$ ✓

$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{18}{30} = \frac{3}{5}$

34 [Intermediate] • Target: Complex numbers: arithmetic

A 30-60-90 right triangle has a short leg of length 4. What is the length of the long leg?

a. 8

b. $4\sqrt{2}$

c. $4\sqrt{3}$ ✓

d. $5\sqrt{3}$

$\text{long leg} = \text{short leg} \times \sqrt{3} = 4\sqrt{3}$

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35 [Intermediate] • Target: Complex numbers: modulus and conjugates

A 45-45-90 right triangle has legs of length 7. What is the length of the hypotenuse?

a. $8\sqrt{2}$

b. $7\sqrt{2}$ ✓

c. 14

d. $7\sqrt{3}$

$\text{hyp} = \text{leg}\sqrt{2} = 7\sqrt{2}$

36 [Intermediate] • Target: Vectors: magnitude and operations

The diagram shows a triangle ABC, where D is a point on side AB and E is a point on side AC. It is given that DE is parallel to BC. If AD = 6 cm, DB = 4 cm, AE = 9 cm, and DE = 12 cm, find the length of the side BC.

a. 18 cm

b. **20 cm** ✓

c. 24 cm

d. 30 cm

Since DE is parallel to BC, triangle ADE is similar to triangle ABC by the AA similarity criterion (angle A is common, and $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$ as corresponding angles). For similar triangles, the ratio of corresponding sides is equal. Therefore, we have: $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$ We know AD = 6 cm and DB = 4 cm, so AB = AD + DB = 6 + 4 = 10 cm. We are given AE = 9 cm and DE = 12 cm. Let BC = x cm. Using the ratio of corresponding sides.

37 [Intermediate] • Target: Radian measure

What is the dot product of the vectors $\vec{u} = \langle 0, 6 \rangle$ and $\vec{v} = \langle -5, -5 \rangle$?

a. 30

b. -30 ✓

c. -4

d. -25

The dot product of two vectors $\vec{u} = \langle u_1, u_2 \rangle$ and $\vec{v} = \langle v_1, v_2 \rangle$ is defined as: $\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$ Substitute the components: $\vec{u} \cdot \vec{v} = (0)(-5) + (6)(-5) = 0 + (-30) = -30$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: measuring the width of a river • Cross-Lesson: 3.01 Functions & Graphs

If $\tan(\theta) = -\frac{3}{4}$ and $\sin(\theta) > 0$, what is the exact value of $\cos(\theta) - \sin(\theta)$?

a. $-\frac{7}{5}$ ✓

b. $-\frac{1}{5}$

c. $\frac{1}{5}$

d. $\frac{7}{5}$

Since $\tan(\theta)0$, θ is in Quadrant II. Here $x = -4, y = 3, r = 5$. Thus $\cos(\theta) = -\frac{4}{5}$ and $\sin(\theta) = \frac{3}{5}$. Their difference is $-\frac{4}{5} - \frac{3}{5} = -\frac{7}{5}$.

39 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the magnitude of the vector $\vec{v} = \langle 9, 40 \rangle$?

a. 42

b. 49

c. **41** ✓

d. 40

$|\vec{v}| = \sqrt{9^2 + 40^2} = \sqrt{1681} = 41$

40 [Challenging] • Target: The Pythagorean identity • Cross-Lesson: 5.01 Statistics & Probability

In the diagram below, ΔABC is inscribed in a circle. AB is a diameter. The length of chord AC is 12 units and the angle $\angle ABC = \theta$. If $\sin(\theta) = \frac{3}{5}$, find the length of the diameter AB.

a. 15 units

b. **20 units** ✓

c. 25 units

d. 30 units

Since AB is the diameter of the circle and C is a point on the circumference, the angle subtended by the diameter at the circumference is a right angle. Thus, $\angle ACB = 90^\circ$. In the right-angled triangle ΔABC, we are given $AC = 12$ and $\sin(\theta) = \frac{3}{5}$, where $\theta = \angle ABC$. In a right-angled triangle, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$. Here, the side opposite to θ is AC and the hypotenuse is AB (the diameter). So.

41 [Challenging] • Target: Exact trigonometric values • Cross-Lesson: 6.01 Geometry

The terminal side of an angle θ in standard position passes through the point $(-3, 4)$. What is $\sin(\theta)$?

a. $\frac{4}{5}$ ✓

b. $-\frac{3}{5}$

c. $-\frac{4}{3}$

d. $\frac{3}{5}$

The distance from origin is $r = \sqrt{(-3)^2 + 4^2} = 5$. Therefore $\sin(\theta) = \frac{y}{r} = \frac{4}{5}$.

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42 [Challenging] • Target: Solving for missing sides and angles • Cross-Lesson: 8.01 Exponentials & Logarithms

If $\sin(\theta) = \frac{7}{25}$ and θ is acute, what is $\cos(\theta)$?

a. $\frac{25}{25}$

b. $\frac{25}{24}$

c. $\frac{7}{25}$

d. $\frac{24}{25}$ ✓

$\cos^2(\theta) = 1 - \sin^2(\theta) = 1 - \left(\frac{7}{25}\right)^2 \Rightarrow \cos(\theta) = \frac{24}{25}$

43 [Challenging] • Target: Complex numbers: arithmetic • Cross-Lesson: 1.01 Heart of Algebra

Convert an angle of 45° into radians in simplest terms.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{\pi}{2}$

Multiply by $\frac{\pi}{180^\circ}$: $45 \times \frac{\pi}{180} = \frac{\pi}{4}$.

44 [Challenging] • Target: Complex numbers: modulus and conjugates • Cross-Lesson: 2.01 Quadratics & Polynomials

A right triangle has legs of length $5\sqrt{3}$ and 5. What is the measure of the smallest angle in degrees?

a. 30° ✓

b. 45°

c. 60°

d. 15°

Tangent of smallest angle = $5/(5\sqrt{3}) = 1/\sqrt{3} \Rightarrow 30^\circ$.

45 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 3.01 Functions & Graphs

The three interior angles of a triangle measure 45°, 65°, and x°. What is the value of x?

a. 70 ✓

b. 80

c. 60

d. 75

Applying geometric definitions gives 70.

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